# Identical Particles

$\rightarrow |\psi_{\alpha\beta}\rangle = |\alpha\rangle_1 \otimes |\beta\rangle_2$ product state

$$P_{12}|\psi_{\alpha\beta}\rangle = |\psi_{\beta\alpha}\rangle = |\beta\rangle_1 \otimes |\alpha\rangle_2$$

$$P_{12}^2 = 1 \rightarrow \pm 1 \text{ eigenvalues.}$$

Projectors: $P_\pm = \dfrac{1 \pm P_{12}}{2}$ ; $P_\pm^2 = P_\pm$

Fermions $\rightarrow P_{12}|\psi_{\alpha\beta}\rangle_- = -|\psi_{\alpha\beta}\rangle \rightarrow$ half Integer

Bosons $\rightarrow P_{12}|\psi_{\alpha\beta}\rangle_+ = +|\psi_{\alpha\beta}\rangle_+ \rightarrow$ Integer

Postulate of spin-statistics.

$$P_{12}|\psi_{\alpha\beta}\rangle_- = \frac{P_{12}}{\sqrt{2}}\left(|\alpha\rangle_1|\beta\rangle_2 - |\beta\rangle_1|\alpha\rangle_2\right)$$
$$= \frac{|\beta\rangle_1|\alpha\rangle_2 - |\alpha\rangle_1|\beta\rangle_2}{\sqrt{2}}$$
$$= -|\psi_{\alpha\beta}\rangle_-$$

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"Leptonium" $\rightarrow l_1^+ l_2^-$ $\quad l_i = \{e, \mu\}$

$$\chi_{0,0} = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle_-|\downarrow\rangle_+ - |\downarrow\rangle_-|\uparrow\rangle_+\right)$$

$$\chi_{1,0} = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle_e|\downarrow\rangle_\mu + |\downarrow\rangle_e|\uparrow\rangle_\mu\right)$$

$P_{e\mu}$? $\qquad \cancel{C}\ \ e^+e^-$ $\qquad C(\chi_{00}) = \chi_{00}$

$C(\chi_{10}) = -\chi_{10}$

$C(\gamma) = -1$

$\Rightarrow (e^+e^-)_0 \rightarrow$ even # of $\gamma$

$(e^+e^-)_1 \rightarrow$ odd # of $\gamma$.

# Molecular Hydrogen $H_2$

$(p^+e^-)(p^+e^-) \to$ electrons

$$\chi_{0,0} = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle_1 |\downarrow\rangle_2 - |\downarrow\rangle_1 |\uparrow\rangle_2\right)$$

$$P_{12}(\chi_{0,0}) = \underline{-\chi_{00}} \Rightarrow \psi_{nlm}(\vec{r}) = +\psi_{nlm}(-\vec{r})$$

$$P_{12}(\underline{\underline{\chi_{1,0}}}) = +\chi_{10} \Rightarrow \psi_{nlm}(\vec{r}) = -\psi_{nlm}(-\vec{r})$$

3:1

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# Molecular Deuterium $D_2$

6:3

Vector $\longrightarrow$ Spin-1

$$3 \times 3 = 1 \oplus 3 \oplus 5 \;/\; 1 \otimes 1 = 0 \oplus 1 \oplus 2$$

$$|0,0\rangle = \frac{1}{\sqrt{3}}\left(|1\rangle_1 |-1\rangle_2 - |0\rangle_1 |0\rangle_2 + |-1\rangle_1 |1\rangle_2\right)$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle_1 |-1\rangle_2 - |-1\rangle_1 |1\rangle_2\right)$$

$$|2,0\rangle = \frac{1}{\sqrt{6}}\left(|1\rangle_1 |-1\rangle_2 + 2|0\rangle_1 |0\rangle_2 + |-1\rangle_1 |1\rangle_2\right)$$

$P_{12}|0,0\rangle \longrightarrow$ Symmetric $\to \underline{\psi(\vec{r}) = +\psi(-\vec{r})}$

$P_{12}|1,0\rangle \longrightarrow$ Anti-symmetric $\psi(\vec{r}) = -\psi(-\vec{r})$

$P_{12}|2,0\rangle \longrightarrow$ Symmetric.

# Molecular Helium $\underline{\underline{{}^4He_2}}$

# Three Fermion systems:

$P_{12}, P_{13}, P_{23} \longrightarrow \underline{\underline{S_3}}$

$|\uparrow\uparrow\downarrow\rangle$

$2 \otimes 2 \otimes 2 = 4_S \oplus 2_M \oplus 2_M$

$\Rightarrow$ No Anti-symmetric piece

$$\Rightarrow \begin{cases} 2 \otimes 2 = 1_A \oplus 3_S \\ 3 \otimes 2 = 4 \oplus 2 = 4_S \oplus 2_M \\ 1_A \otimes 2 = 2_M \end{cases}$$

Totally Anti-symmetric state?

$$|abc\rangle_A = \frac{1}{\sqrt{6}} \sum_\sigma \mathrm{sign}(\sigma)\, |\sigma(a)\sigma(b)\sigma(c)\rangle$$

$$= \frac{1}{\sqrt{6}} \Big( |abc\rangle - |acb\rangle + |bca\rangle - |bac\rangle + |cab\rangle - |cba\rangle \Big)$$

$\rightarrow$ No such thing if $b = c$.

$\rightarrow |\Delta^{++}\rangle = \varepsilon_{abc} |u_a^\uparrow u_b^\uparrow u_c^\uparrow\rangle \rightarrow$ "color" Strong Interaction

Spin $3/2$ $\rightarrow$ Fermion

triton $^3H \rightarrow |nnp\rangle$

~~$|nnn\rangle$~~

# Bosonic enhancements

$$|\alpha\rangle_1 \otimes |\beta\rangle_2 \longrightarrow \underbrace{|c\rangle \otimes |c\rangle}_{\text{Symmetric}}$$

$$\langle c|\hat{U}|\alpha\rangle = t_{\alpha c}$$

$$\langle c|\hat{U}|\beta\rangle = t_{\beta c}$$

**Distinguishable:** $\langle c| \otimes \langle c| (\hat{U} \otimes \hat{U}) |\alpha\rangle_1 |\beta\rangle_2$

$$t_{\alpha\beta\to cc} = t_{\alpha c}\, t_{\beta c} \Rightarrow \Gamma_{DIST} = |t_{\alpha c}|^2 |t_{\beta c}|^2 = \Gamma_{\alpha c} \times \Gamma_{\beta c}$$

**Two Identical Bosons**

$$|i\rangle_+ = \frac{\left(|\alpha\rangle|\beta\rangle + |\beta\rangle|\alpha\rangle\right)}{\sqrt{2}}$$

$$|f\rangle_+ = |c\rangle_1 |c\rangle_2$$

$$t_{\alpha\beta\to cc} = \frac{\left(t_{\alpha c} t_{\beta c} + t_{\beta c} t_{\alpha c}\right)}{\sqrt{2}} = \sqrt{2}\, t_{\alpha c} t_{\beta c}$$

$$\Gamma_{BOSONIC} = 2 \times \Gamma_{\alpha\to c} \Gamma_{\beta\to c}$$

Bose-Einstein
Stimulated Emission
etc.

$$\Gamma^{(N)}_{BOSONIC} = N!\, \Gamma^{(N)}_{DIST}$$